THE HONG KONG POLYTECHNIC UNIVERSITY
香港理工大學

BSE4610-

Building Informatics

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BSE4610 Building Informatics

Week 1: Subject Briefing & Introduction

Today's focus:

- Course overview and expectations
- What is Building Informatics?
- Building data ecosystem

Class format:

Interactive lecture + short activities; laptops welcome

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Any time ☺
ZS831



Prof. Zhiling Guo (ZLG)
Department of Building Environment and Energy Engineering

Research interests:
- Machine Learning,
- Geoscience,
- AIoT,
- Renewable Energy Technologies.

Session Agenda and Time Plan

Mini-session 1: Course Overview ~50 min

- Aims, structure, learning outcomes (LOs)
- Assessment scheme and expectations
- Background pulse-check

Mini-session 2: What is Building Informatics? ~50 min

- Definitions, scope, value chain
- Role in smart buildings and industry

Mini-session 3: Building Data Ecosystem ~40 min

- Sources, data types, lifecycle from acquisition to action

Q&A and wrap-up ~10 min

Total: 150 minutes (2 hour 30 min)

Housekeeping and Logistics

Format

L+T (3 hours) with slides, discussions, quick polls

Materials

Canvas slides, readings, announcements, submissions

Tools

- Python/Jupyter (basic), Excel/Power BI (visualization)
- Optional: MQTT sandbox/BIM viewer
- Complete environment by Week 3

Communication

Email, Canvas, office hours

Etiquette

- Punctuality, active participation
- Devices on silent

Academic Integrity

- Cite sources; reproducible analysis
- Disclose any generative AI assistance

Why Building Informatics Now?

Industry Drivers

- Decarbonization and net-zero commitments
- Cost, reliability, and grid interaction
- Health, safety, and compliance
- Digital transformation of operations

Technology Enablers

- IoT sensors and gateways; cloud computing
- Open APIs and data standards
- ML/AI for time-series and decision support

Paradigm Shift

From **"data-rich, insight-poor"** to **"data-informed action"**

Career Relevance

Hybrid BSE + data skills are in **high demand**

Quick question: "Name one building decision that could be improved if we had better data. What data would you ask for?"

Course Aims and What You Will Gain

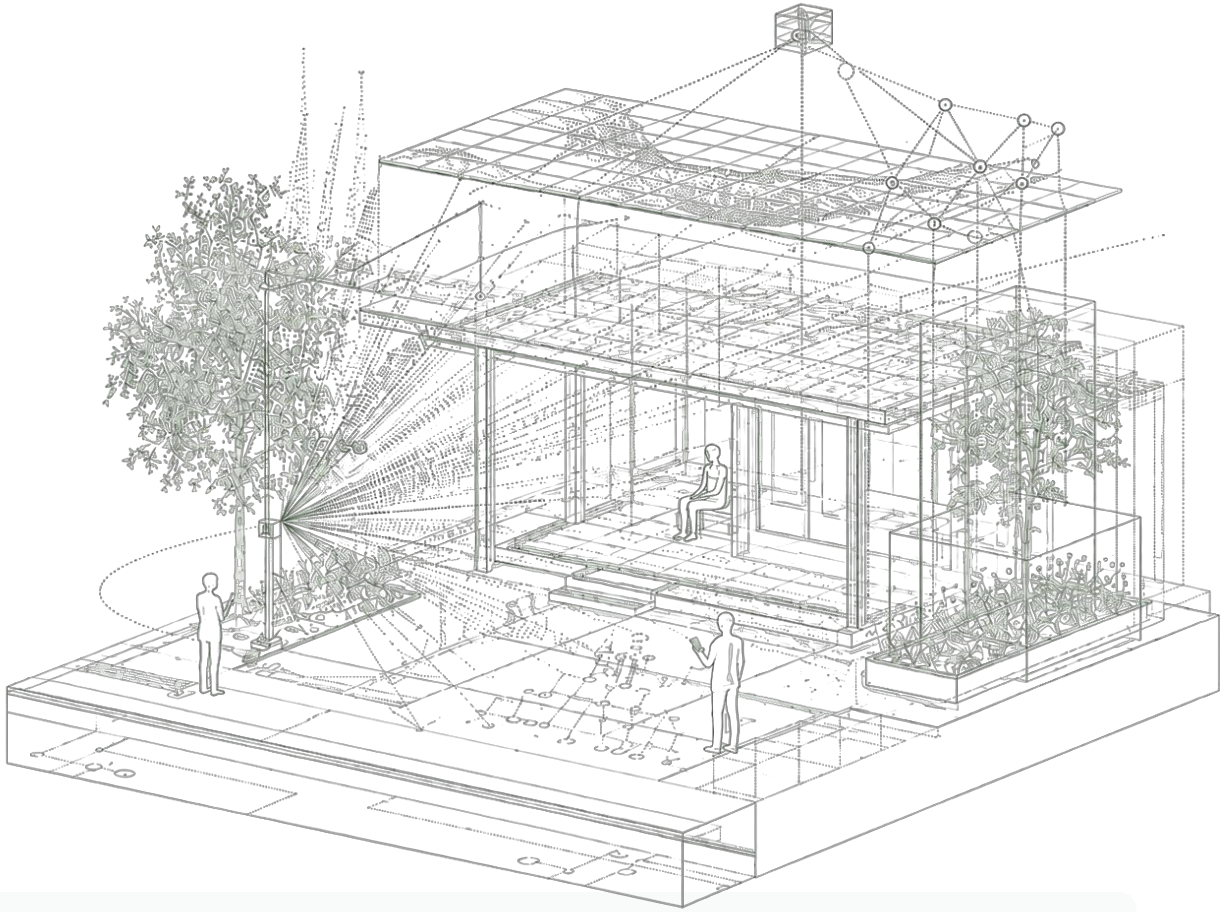
Build foundations
Concepts, systems, and data flow in Building Informatics

Understand IoT and AI applications
Strengths, limits, deployment considerations

Practice basic ML
Hands-on with building datasets and simple models

Translate insights into actions
Efficiency, IEQ, safety, and operations

Collaborate across disciplines
Communicate with IT/AI professionals



Which aim feels most challenging to you right now, and why?

Course Structure and Weekly Flow

Weeks

Foundations

Building Automation Systems, Internet of Things, Machine Learning, AI

Week 8

General Holiday

No class scheduled

Week 5

Mini Project Workshop

Team formation, data selection, workflow planning

Week 9-11

Applications

Energy optimization, Indoor Environmental Quality & Safety, Standards/Integration

Week 12-13

Seminar Presentation (1 Group)

25-minute presentations + 5-minute Q&A sessions

Pedagogy Approach

- Lecture + demonstration + case studies
- Continuous feedback on progress
- Hands-on activities and participation
- Application-focused learning

Course Structure and Weekly Flow

Week 1 – Subject Briefing & Introduction to Building Informatics

Mini-session 1: Course Overview

- Course aims, structure, and learning outcomes
- Assessment scheme and project expectations
- Student background survey

Mini-session 2: What is Building Informatics?

- Definitions and scope
- Role in smart buildings and industry transformation
- Interdisciplinary nature (BSE + IT + AI)

Mini-session 3: Building Data Ecosystem

- Sources: sensors, BAS, IoT, BIM
- Data types: structured, unstructured, time-series
- Data lifecycle: acquisition → processing → application

Course Structure and Weekly Flow

Week 2 – Technical Foundations of Building Informatics

Mini-session 1: Building Automation Systems (BAS)

- Overview of HVAC, lighting, security systems
- Control logic and data flow
- BAS architecture and communication protocols

Mini-session 2: IT Infrastructure & Data Acquisition

- Networks: LAN, Wi-Fi, Zigbee, BACnet/IP
- Middleware platforms and cloud services
- Cybersecurity basics in buildings
- Time-series data collection and storage

Mini-session 3: Introduction to BIM in Building Informatics

- What is BIM? Basic concepts and tools (Revit, ArchiCAD)
- BIM vs. CAD; structured data and metadata
- BIM as a digital model for building systems
- Use case: BIM + IoT for Digital Twin

Course Structure and Weekly Flow

Week 3 – Machine Learning Basics for Building Data

Mini-session 1: Introduction to Machine Learning

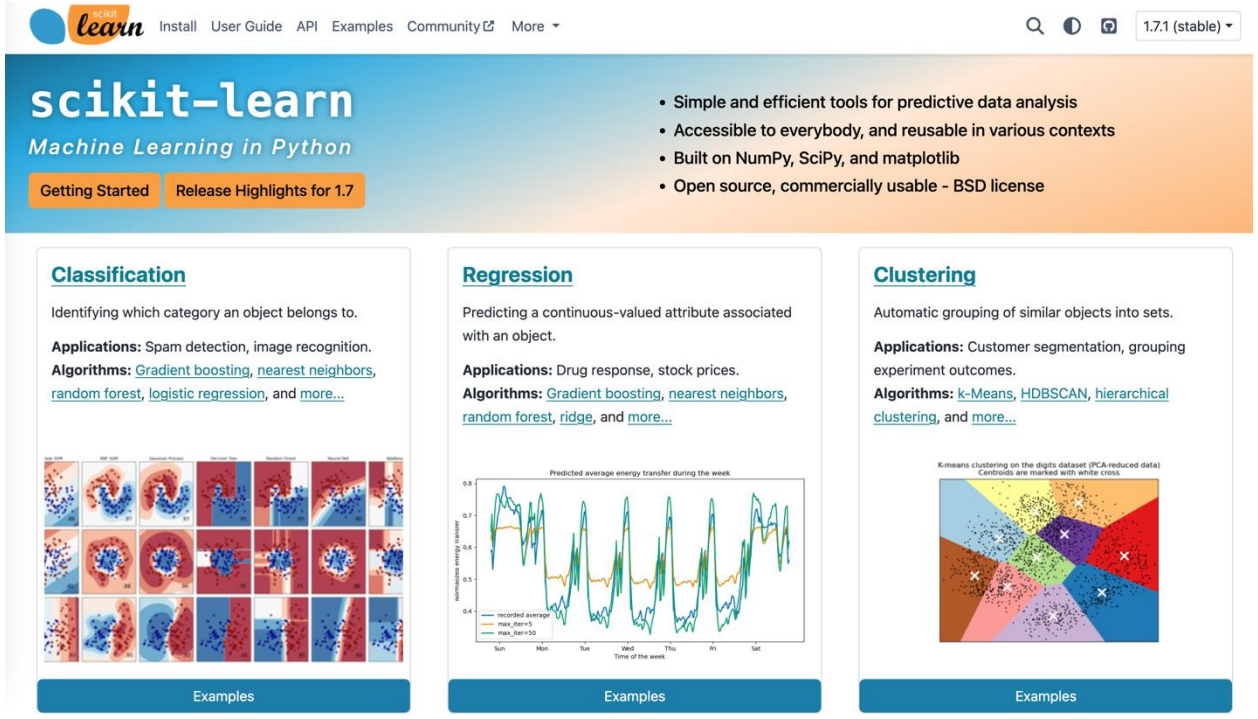
- Supervised vs. unsupervised learning
- Features, labels, and datasets
- Common algorithms: linear regression, K-means, etc.

Mini-session 2: Data Preprocessing

- Cleaning, normalization, feature selection
- Handling missing/incomplete building data
- Tools: Python, Jupyter, pandas

Mini-session 3: ML Hands-on Demo

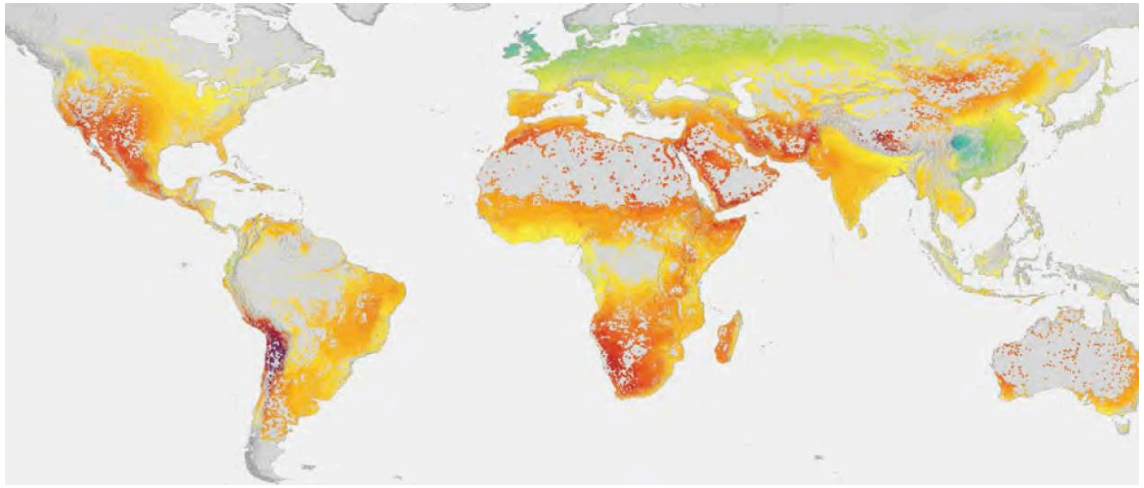
- Load a simple dataset
- Train a linear regression model to predict energy use
- Visualize prediction vs. actual

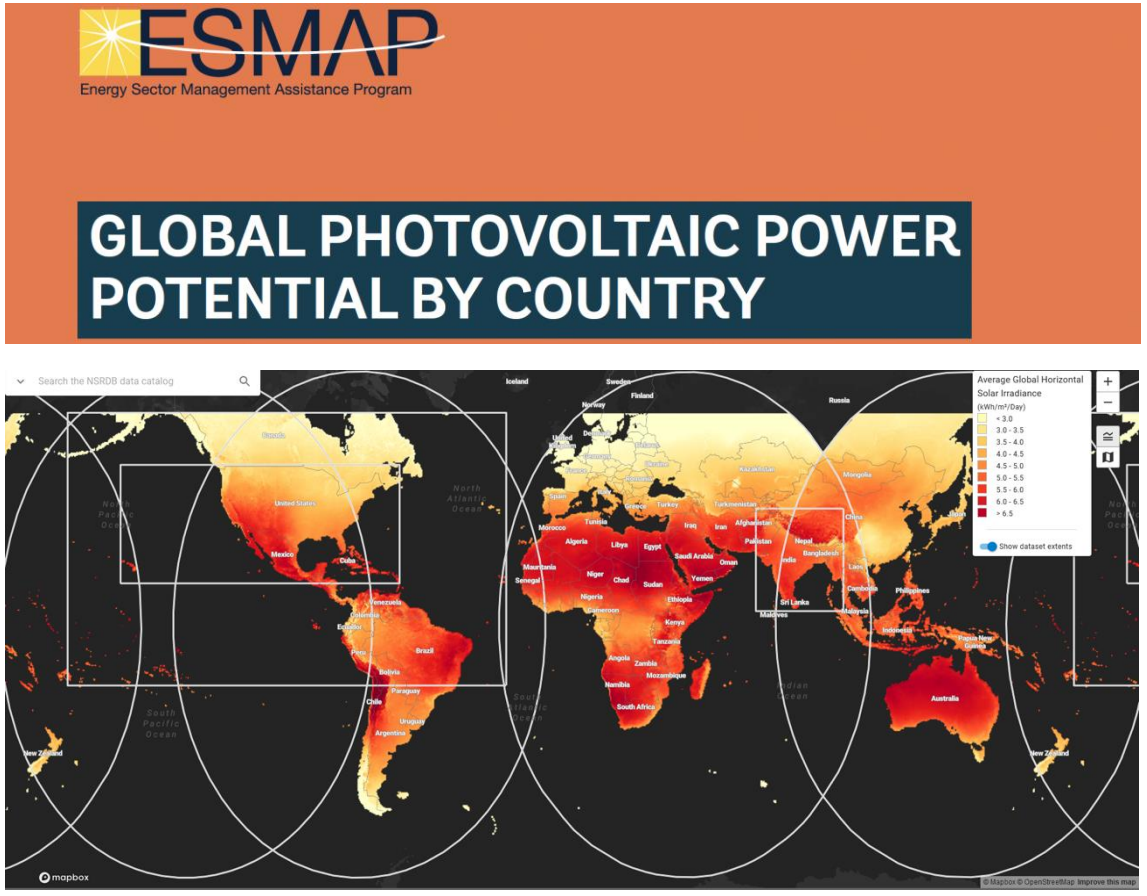


https://scikit-learn.org/stable/

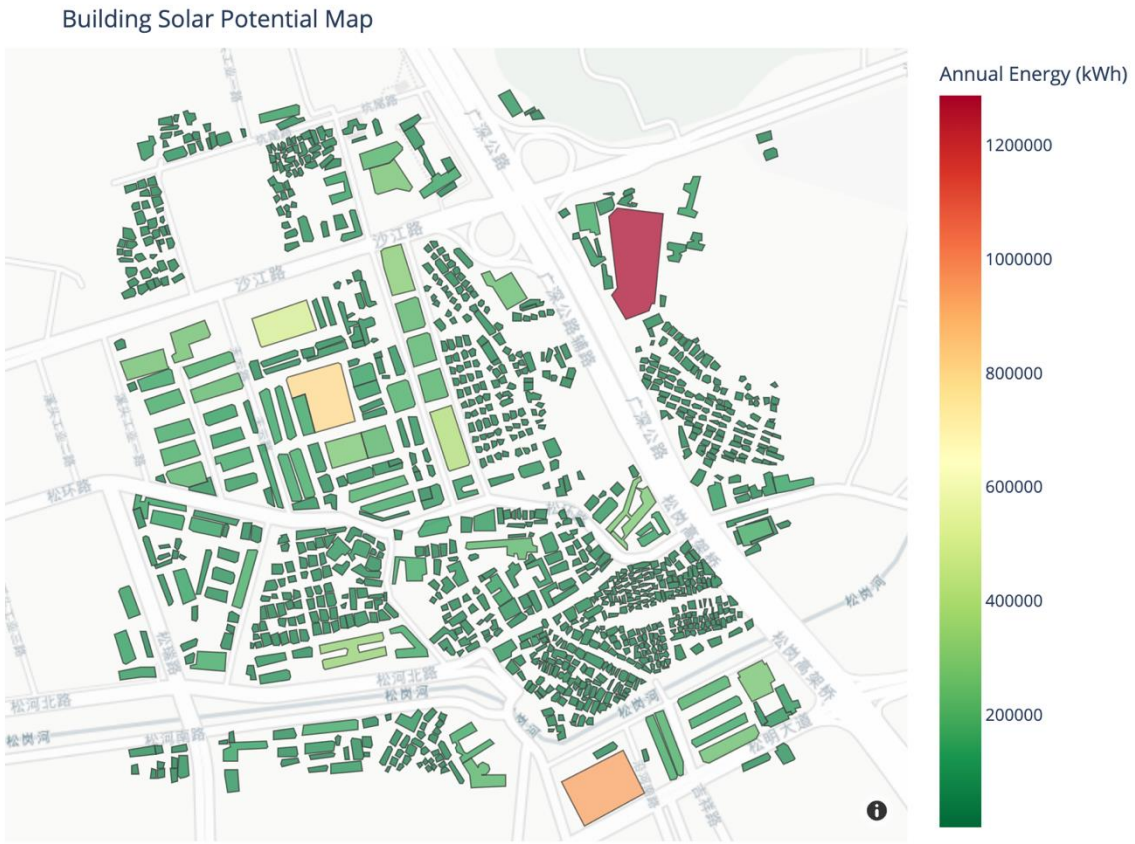
Course Structure and Weekly Flow

Week 4 – PV Assessment for Building Informatics





Course Structure and Weekly Flow



Week 5 – Mini Project Workshop

Mini-session 1: Project Launch

- Mini project themes and objectives
- Teaming and topic confirmation
- Project rubric and deliverables

Mini-session 2: Methodology & Tools

- Suggested data sources (open datasets or simulated data)
- Tools: Python, Excel, Power BI
- Suggested workflow: problem → data → model → insight

Mini-session 3: Group Work & Consultation

- Students begin planning
- Instructor provides group-specific feedback
- Q&A and clarification

Course Structure and Weekly Flow

Week 6 – Advanced AI Applications in Buildings

Mini-session 1: Deep Learning in Building Informatics

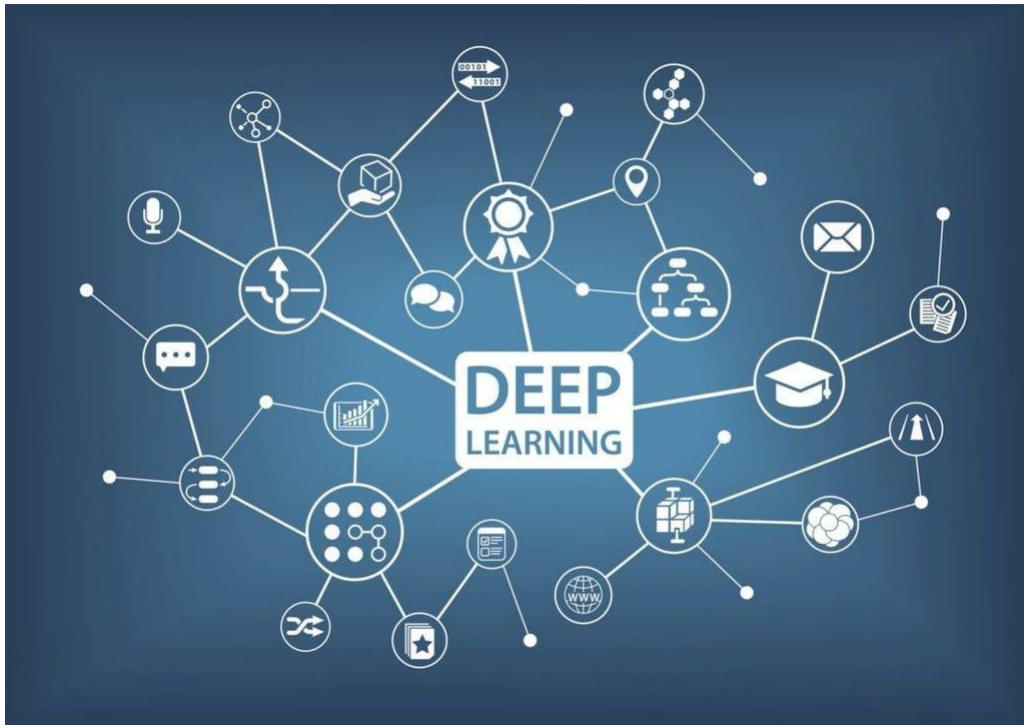
- Neural networks basics
- Time-series prediction (LSTM, RNN, Transformer)
- Vision-based applications (e.g., occupancy via cameras)
- Recent Large Language Models (LLM)

Mini-session 2: AI for Decision Support

- Fault detection & diagnostics
- Energy optimization recommendations
- Integration with BMS

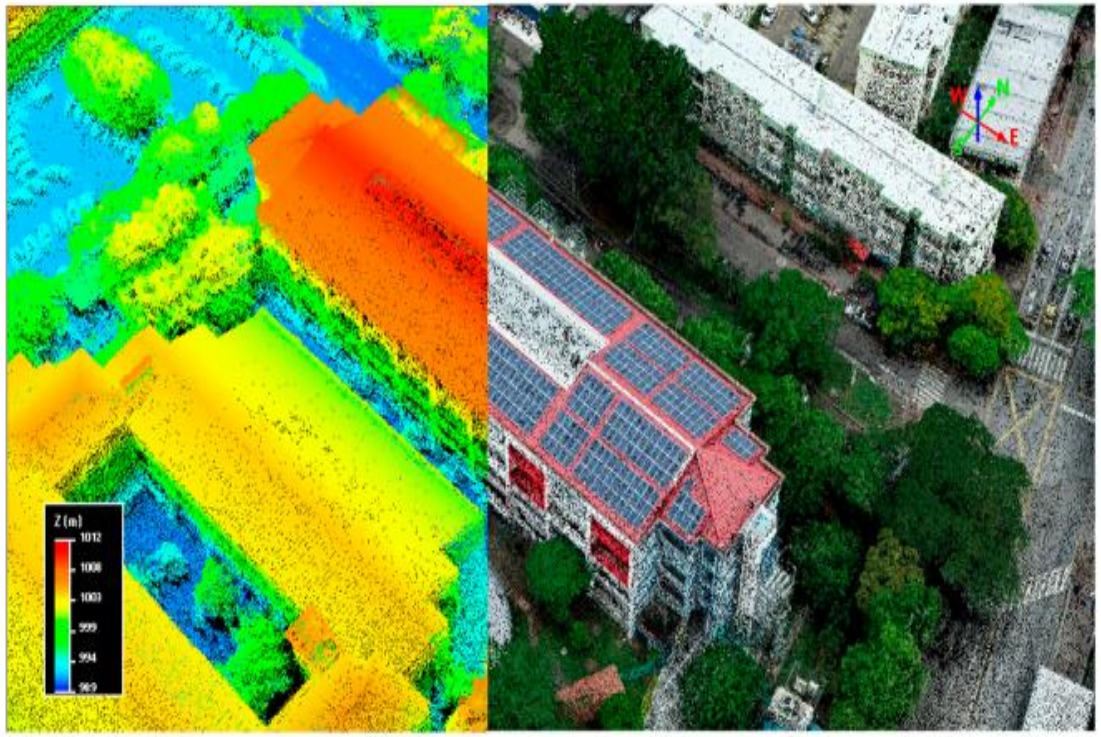
Mini-session 3: Case Study & Discussion

- AI-enabled smart building platform
- Analyze strengths, weaknesses, deployment challenges
- Discuss ethical & privacy concerns



https://medium.com/hackernoon/introduction-to-deep-learning-9064d6b87a51

Course Structure and Weekly Flow



Week 7 – Advanced AI Applications in Buildings

(supply side)

- Brief Introduction to Remote Sensing (RS)
- PV Potential Estimation
- PV Array Detection
- PV Fault Monitoring & Diagnosis
- Generative AI for Renewable Energy

Course Structure and Weekly Flow

Week 9 – IoT Technologies in Buildings

Mini-session 1: IoT Fundamentals

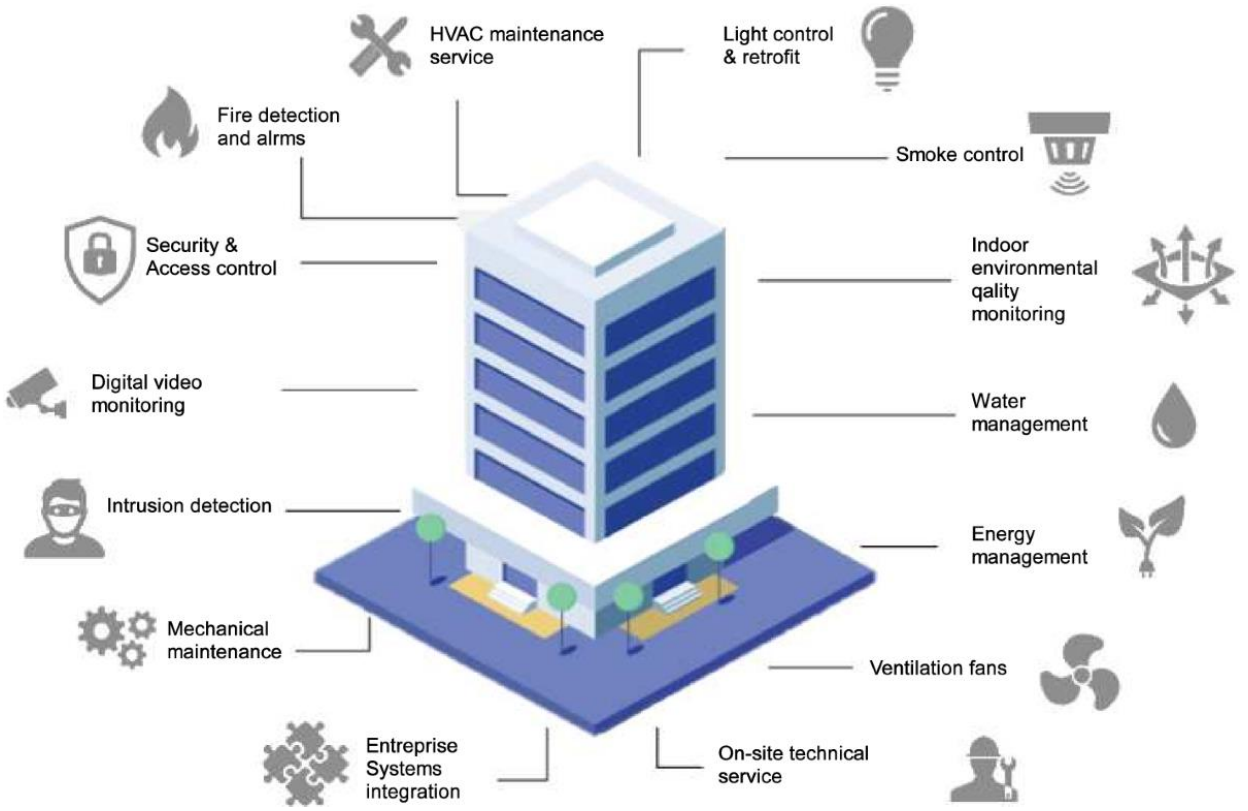
- IoT architecture and protocols (MQTT, CoAP)
- IoT device types (sensors, actuators, controllers)
- Real-time data transmission

Mini-session 2: Applications of IoT in Buildings

- Smart HVAC, occupancy detection, lighting
- Predictive maintenance and asset monitoring
- Case study walkthrough

Mini-session 3: IoT System Design Activity

- Group task: Design a basic IoT system for a smart room
- Define sensors, data flow, and use cases
- Peer sharing and feedback



https://link.springer.com/article/10.1007/s10462-022-10286-2

Course Structure and Weekly Flow

Week 10 – Built Environment Quality & Safety

Mini-session 1: Indoor Environmental Quality (IEQ) Monitoring

- Parameters: temperature, CO_2, PM2.5, VOCs
- Comfort vs. health trade-offs
- Sensor calibration & placement

Mini-session 2: Safety & Anomaly Detection

- Incident detection via video/image/thermal sensors
- Anomaly detection algorithms
- Real-time alert systems

Mini-session 3: Group Case Discussion

- Analyze a failure incident (e.g., overheating, intrusion)
- Identify causes and propose a data-driven solution
- Written assignment



https://www.istockphoto.com/photos/indoor-fireplace-smoke

Course Structure and Weekly Flow

Week – Energy Efficiency & Cross-Disciplinary Collaboration (Optional)

Mini-session 1: Energy Data & Optimization Strategies

- Energy use intensity (EUI), load profiles
- Peak demand analysis
- Occupancy-linked energy patterns
- Data-driven HVAC control
- Demand response techniques, predictive control

Mini-session 2.1: Working with IT/AI Professionals

- Role of domain experts vs. data scientists
- Common communication gaps
- Translating engineering problems into data problems

Mini-session 2.2: Industry Standards & Integration

- BIM, BACnet, IFC, OPC UA
- Data interoperability challenges
- Smart city context and frameworks

Mini-session 3: Future Trends & Career Pathways

- Digital twins, generative AI, cloud-based BMS
- Emerging roles in building informatics
- Reflections: What's next for smart buildings?



https://www.nature.com/articles/s41893-018-0141-x

Contact Hours and Expected Effort

Contact Hours

39 hours total:

- Lectures & Tutorials: 27 hours
- Seminars: 6 hours
- Mini-Project: 3 hours

Total Student Effort

~120 hours including self-study and project work

Suggested Weekly Rhythm

- Pre-read/recap: 1–2 hours
- Hands-on practice: 1–2 hours
- Project work (avg): 1–2 hours



Learning Outcomes (LOs) and Evidence

LO **Understand technical foundations of Building Informatics**

Evidence: Precise definitions; system and data-flow diagrams; protocol comparisons

LO **Know typical IoT and AI applications in buildings**

Evidence: Use-case critique; benefit–risk analysis; deployment constraints

LO **Basic ML skills for building data**

Evidence: Runnable notebook; metrics (e.g., RMSE); plots with clear labels

LO **Apply insights to efficiency, IEQ, safety**

Evidence: KPI dashboard; scenario analysis; actionable recommendations

LO **Communicate with IT/AI professionals**

Evidence: Problem statement; data schema; assumptions and handoff document

How LOs Map to Weeks and Activities

Assessment Alignment

Written Assignment (15%)

LO1 LO2

Synthesize foundations and applications with clear structure and proper citations

Mini Project (20%)

LO3 LO4 LO5

Problem → data → model → insight → recommendation; reproducible artifacts

Examination (50%)

LO1 LO2 LO3 LO4

Conceptual + applied reasoning; case/data-oriented questions

Assessment Scheme Overview (50% Coursework + 50% Exam)

Coursework 50%

Written assignment 15%

- Synthesize foundations; proper citations

Seminar presentation 15%

- 24 min talk + 6 min Q&A; technical rigor

Mini project 20%

- Problem → data → model → recommendation
- Reproducible artifacts

Examination 50%

- Conceptual + applied reasoning
- Case/data-oriented questions

Expectations and policies

- Academic integrity; cite sources; disclose AI use
- Reproducibility: code, data, version control
- Teamwork: defined roles; fair contribution

Examination (50%): Format and Focus

Focus Areas

- Concepts: AI, IoT, building data, lifecycle, standards
- Applied reasoning: interpret plots, diagnose issues, propose actions

Question Types

- Multiple choice questions
- Short answers, figure/plot interpretation
- Case-based scenarios with data snippets
- Brief calculations or comparisons

Materials and Logistics

- Duration, allowed materials, and detailed scope will be confirmed by Week 9
- Calculators permitted unless stated otherwise

Scoring Principles

- Clarity of reasoning earns partial credit
- Use correct units and clearly state assumptions

Written Assignment (15%): Scope and Rubric Highlights

Purpose

Synthesize foundations; analyze an application or integration topic

Possible Themes

- BAS data flows and control logic
- Protocol comparison (e.g., BACnet vs. MQTT roles)
- BIM–IoT integration for a digital twin
- ML/AI

Expected Structure

- Executive summary (150–200 words)
- Background and definitions
- Analysis with evidence/figures
- Discussion: limits, risks, standards
- Conclusion and references

Rubric Highlights (100%)

- **30%** Technical accuracy
- **25%** Critical analysis and depth
- **20%** Use of evidence/figures and citations
- **15%** Organization and clarity
- **10%** Professional presentation

Seminar Presentation (15%): Expectations and Delivery

Format

- 24min presentation + 6min Q&A (Weeks 12–13)
- Audience: technical peers; aim for clarity + rigor

3 people per group ✅

Slide Guidance

- Problem statement and context
- Data: sources, quality, preprocessing
- Method: why this approach; key parameters
- Results: plots/tables; key findings
- Recommendations: actionable, measurable, risks

Rubric Highlights (100%)

- Technical content and correctness **(35%)**
- Clarity of narrative and structure **(20%)**
- Timing and delivery **(10%)**
- Team coordination **(5%)**
- Visualization quality and readability **(15%)**
- Q&A handling and depth **(15%)**

Mini Project (20%): End-to-End Pipeline

Objective

Solve a building-relevant problem using data

3 people per group ✅

Workflow

Problem Framing — Data Acquisition — Processing — Modeling — Insight — Recommendation

Deliverables

- 8–12 slides
- Short report or extended slide notes (4–8 pages)
- Reproducible artifacts: data sample, code/notebook, README

Evaluation (100%)

- Problem framing and goals 15%
- Data pipeline and quality management 20%
- Methodology choice and justification 20%
- Results and validation 15%
- Insights and actionable recommendations 15%
- Reproducibility and documentation 10%
- Teamwork and project management 5%

Tools and Software: Setup and Minimum Skills

Core Analytics Stack

- Python 3.x, JupyterLab/Notebook
- Key libraries: pandas, numpy, matplotlib/seaborn, scikit- learn

ANACONDA.

spyder

https://www.anaconda.com/download/success

Visualization

Script, Power BI or Excel (charts, pivots, slicers)

Setup Guidance

- Install by end of **Week 3**
- Verify with a simple notebook (import libs; read CSV; plot)

Optional for IoT/BIM

- MQTT sandbox/testing tool
- Basic IFC/BIM viewer

Good Practice

- Version control (Git or dated folders)
- Commented notebooks; consistent file

Data Sources for Coursework: Options and Ethics

Potential Sources

- Provided sample datasets (energy, IEQ, equipment logs)
- Public datasets (building energy/time-series; no PII)
- Simulated or anonymized datasets
- Contextual data: weather, schedules, space metadata

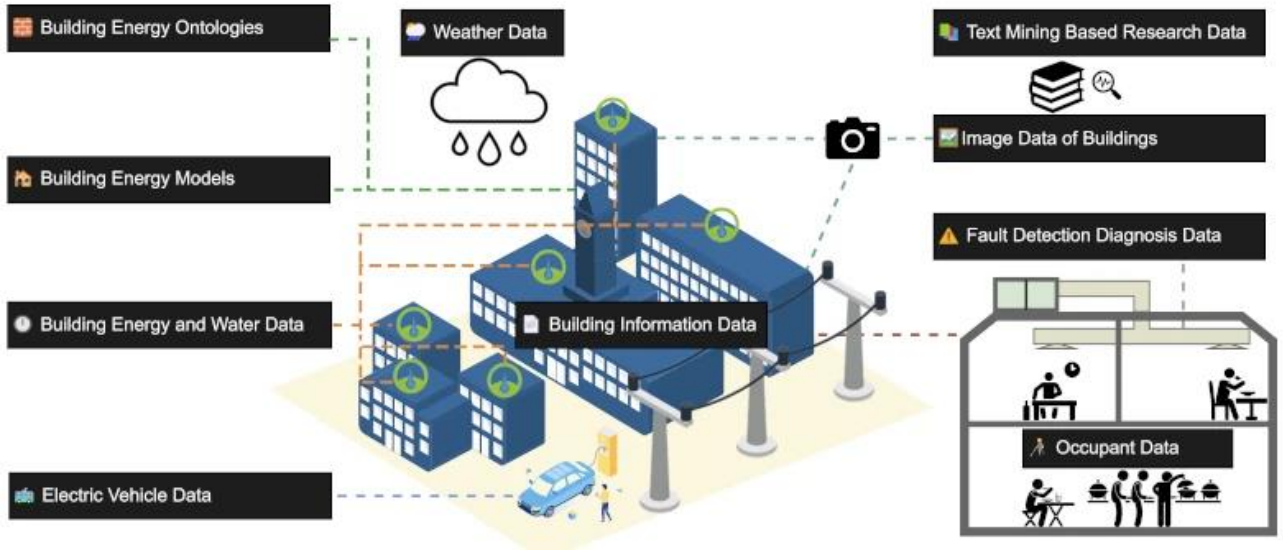


Image source: Buildings.Evolved.

Selection Criteria

- Relevance to problem; data volume/granularity
- Data quality: completeness, units, timestamps
- Rights and ethics: licensing, privacy, consent

Data Source Examples

Sensors BAS Cloud BIM

Documentation

- Data dictionary; units; sampling intervals
- Known issues and cleaning steps

Policies and Expectations: Integrity, AI Use, and Teamwork

Academic Integrity

- Cite all sources; no plagiarism
- Keep analysis reproducible (data, code, steps)

Generative AI

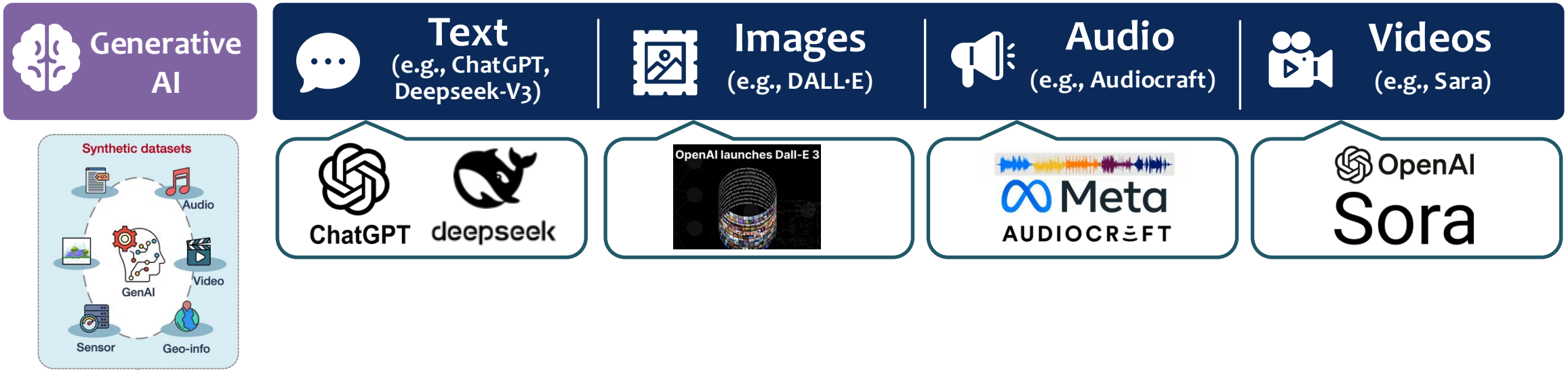
- Permitted as assistive tool; disclose where/how used
- You are responsible for verifying correctness

Teamwork

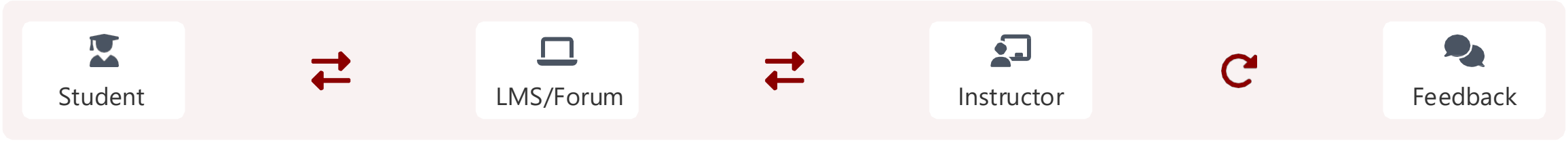
- Define roles; rotate as needed (analyst, viz, presenter)
- Keep a brief team log (decisions, contributions)

Professional Conduct

- Respectful discussion; constructive feedback
- Deadlines and extensions: see Canvas; communicate early



Communication and Feedback: How We Work Together



Channels

- LMS for announcements, materials, submissions
- Email for personal matters; office hours for deep dives
- Discussion forum for shared Q&A

Response and Feedback

- Typical response time: within 3 working days
- Rubric-based feedback for coursework

Your Voice Matters

- Raise risks early; ask questions often
- Suggest topics for demos
- Provide feedback to improve the course

Quick Poll: Your Experience Snapshot

Rate your current level (select one per line):

Programming (Python/Excel scripting): None | Basic | Intermediate | Advanced

BAS or controls exposure: None | Some exposure | Hands-on

IoT (sensors, MQTT, gateways): None | Some exposure | Hands-on

ML/AI (regression, clustering, DL): None | Intro course | Applied projects

Data viz tools (Excel/Power BI): None | Basic | Proficient

BIM (Revit/IFC concepts): None | Basic | Proficient

What we ask

- Prior exposure: BAS, IoT, ML/AI, BIM, visualization
- Programming comfort and toolchains
- Interests and career goals

Quick Poll: Interests and Application Domains

Choose your top two interests:

Energy efficiency and demand response
Optimization, grid interaction, peak reduction

Indoor environmental quality (IEQ) & health
Air quality, thermal comfort, lighting, acoustics

Digital twins (BIM + live data)
Real-time visualization, 3D integration

Smart campus/city integration
Multi-building systems, urban scale applications

Operations &maintenance)
Fault detection, predictive maintenance

Safety, security, and anomaly detection
Critical systems monitoring, pattern detection

Optional: Career goal focus

Engineer | Data analyst | Product/Platform | Research

Icebreaker: Two-Minute Pair Share

Instructions

- Pair up with someone you do not know
- 1 minute each to share:
  - One building technology you admire
  - One concern you have about smart buildings
  - Be ready to share one highlight with the class

Discuss

A standout building or system you visited

"I was impressed by the automated shading system at..."

A frustrating experience with building controls

"The thermostat in my apartment always seems to..."

Person 1: 1 minute | Person 2: 1 minute | Class sharing

Core Reading List with Annotations

Bakker (2019), Smart Buildings
Design and technology integration; human-centered outcomes

Wang S.W. (2009), Intelligent Buildings and Building Automation
Controls fundamentals; sequences, automation logic

Hwang & Chen (2017), Big Data Analytics
Data architecture; analytics patterns relevant to buildings

Sinopoli (2016), Advanced Technology for Smart Buildings
Practical systems view; BAS, integration strategies

McGowan (2015), Energy and Analytics
Data pipelines for energy insights; case studies

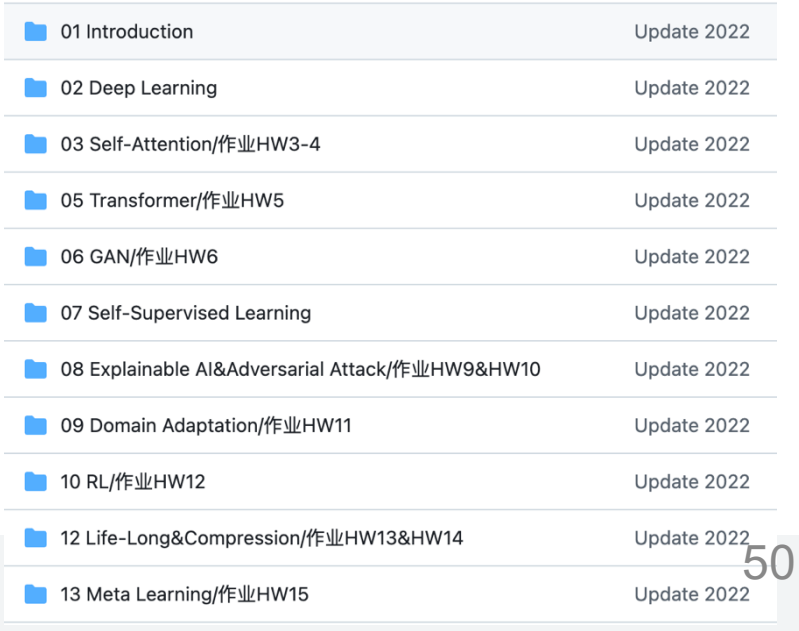
Zhou (2021), Machine Learning
Christopher M. Bishop (2023), Deep Learning
ML theory and practice; reference for Weeks 3–5

Access: See LMS for chapter excerpts and links

For Machine Learning Beginners:

[Lhy_Machine_Learning]

https://github.com/Fafa-DL/Lhy_Machine_Learning/tree/main/2021%20ML



Transition: From Logistics to the Field Itself



We have covered:

- Clarified aims, learning outcomes, and assessment scheme
- Understood tools, policies, and support resources
- Explored study strategies and success factors
- Surveyed backgrounds and interests for tailored learning

Coming up next:

- What is Building Informatics as a discipline?
- Why it matters in modern building engineering
- How it fits with related domains (BAS, BIM, IoT)
- The value chain from data to actionable insights

Learning Objective

Define the field of Building Informatics and differentiate it from related domains (BAS, BIM, FM), recognizing its unique value proposition in the built environment.

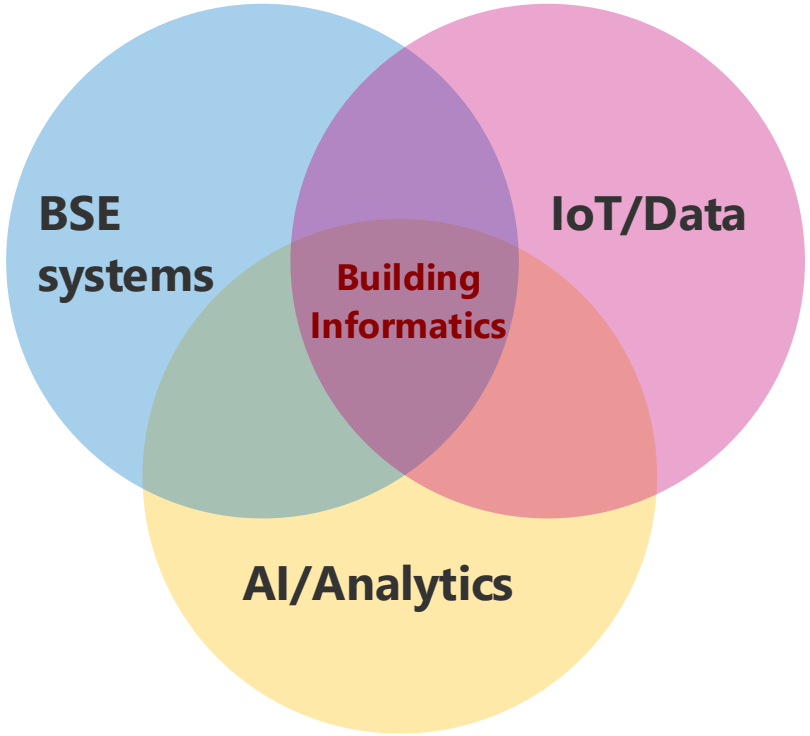
What Is Building Informatics? Definition and Boundaries

Working Definition

*The discipline of **acquiring, structuring, analyzing, and applying building-related data** to improve performance, comfort, safety, and cost across the **building lifecycle**.*

Core Elements

- Systems context: HVAC, lighting, security, meters
- Data context: sensors, BAS, IoT, BIM, external (weather, tariffs)
- Methods: analytics, ML/AI, visualization, decision support
- Action: recommendations, control changes, workflows



≠ Distinctions (not the same as)

- BAS/controls alone (through programmed logic)
- BIM alone (focuses on geometry and structured design data)
- Facility management alone (focuses on processes and work orders)

Scope of Building Informatics: Lifecycle and Value Chain

Lifecycle Coverage

Design
Data standards, simulation inputs, BIM semantics

Construction/Commissioning
Progress/quality data, functional testing

Operations
Monitoring, optimization, FDD, DR

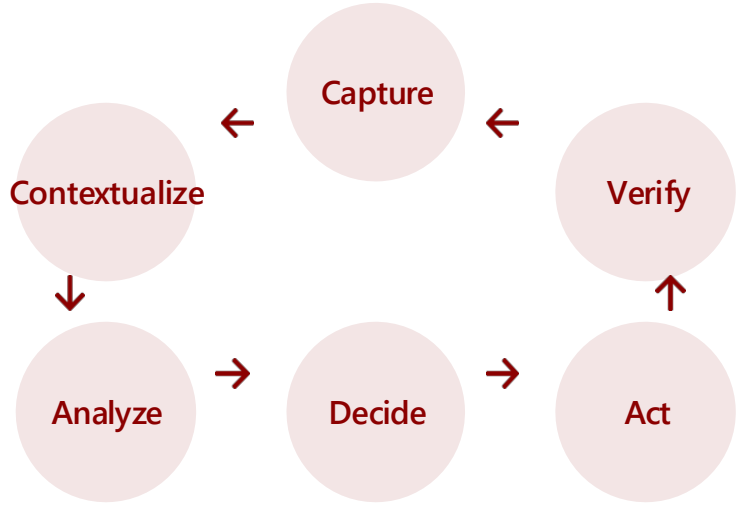
Retrofit/End-of-Life
Performance baselines, Return on Investment (ROI) analytics

Typical Outcomes

- Energy savings; comfort/IEQ improvements
- Fault reduction and faster MTTR
- Compliance and risk mitigation
- Better asset planning and utilization

Value Chain

Capture → Contextualize → Analyze → Decide → Act → Verify



The value chain is cyclical. Verification feeds back into models and rules, improving future performance. This "learning loop" is the hallmark of a mature informatics practice.

KPIs make outcomes visible and measurable: kWh, peak kW, CO_2 ppm, PM2.5 μg/m³, fault counts, response times.

Positioning Among Related Fields

Building Informatics vs. BAS (Controls)

BAS:

- Sequences, setpoints, actuation
- Alarms and notifications

Informatics:

- Data modeling, analytics
- Decision support across systems

Building Informatics vs. BIM

BIM:

- Geometry, components
- Relationships for design/construction

Informatics:

- Links BIM semantics
- Live operational data integration

Building Informatics vs. Facility Management

FM:

- Processes, work orders
- Compliance and operations

Informatics:

- Evidence-based insights
- Optimize FM actions and priorities

Building Informatics vs. Data Science

Data Science:

- Generic methods and platforms
- Generalized analysis techniques

Informatics:

- Domain-specific data and context
- Building constraints and outcomes

Key Questions Building Informatics Answers (Examples)

Performance and targets

Are we meeting energy/IEQ targets? Where and why not?

Data: Energy meters, IEQ sensors → **Method:** Threshold tracking → **Action:** Adjust setpoints

Diagnostics and risk

Which systems show abnormal behavior or pending faults?

Data: Equipment logs, alarms → **Method:** Anomaly detection → **Action:** Prioritize maintenance

Optimization and operations

What should we change (setpoints, schedules)? When and by how much?

Data: BAS, occupancy, weather → **Method:** Predictive control → **Action:** Implement schedule changes

Investment and planning

Which retrofit yields the best ROI under uncertainty?

Data: Energy history, costs, models → **Method:** Scenario analysis → **Action:** Investment decisions

Evolution of Smart Buildings

Control Evolution

- Scheduled → Sensor-based → Rule-based → Model-based → Data-driven/AI-assisted

Integration Evolution

- Siloed systems → Middleware → Platform ecosystems → Digital twins

Infrastructure Evolution

- On-prem BMS → Cloud/edge hybrids → API-first → Microservices

Standards Evolution

- Proprietary → Open protocols (BACnet, Modbus) → Semantic tagging (Haystack, Brick) → Interoperable models (IFC, OPC UA)

Smart buildings have evolved from simple, isolated, and proprietary systems to:

- **Intelligent**
- **Interconnected**
- **Cloud-enabled**
- **Standardized and interoperable**

Industry Drivers: Why Organizations Invest

Sustainability and Regulation

- Net-zero roadmaps
- Carbon pricing
- Disclosure requirements (ESG)

KPI: tCO_2e reduced/year

Cost and Resilience

- Energy price volatility
- Peak demand charges
- Grid services

KPI: Peak kW reduction

Health and Safety

- Post-pandemic IEQ expectations
- Safety compliance
- Occupant comfort

KPI: % time in comfort range

Workforce and Operations

- Skilled labor shortages
- Knowledge capture
- Automation

KPI: Staff hours saved/month

Digital Strategy

- Data as an asset
- Interoperability mandates
- Cybersecurity posture

KPI: % systems with API access

Stakeholders and Roles

Core Stakeholders

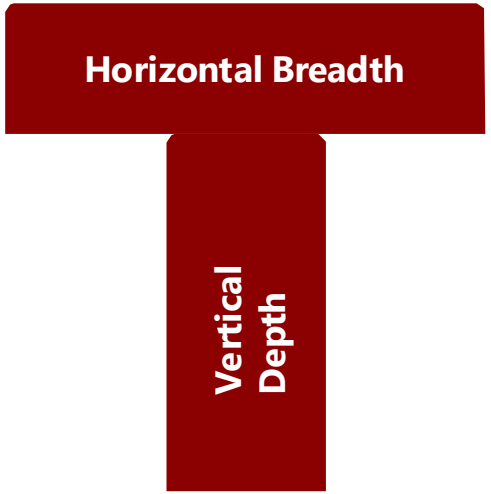
- **Owners/asset managers:** strategy, ROI, risk
- **Facility managers:** daily control, alarms, work orders
- **Energy/IEQ managers:** KPIs, optimization, reporting
- **Engineers (BSE):** systems expertise, sequences, constraints
- **IT/security:** networks, identity, cybersecurity, data governance
- **Data scientists/analysts:** modeling, dashboards, insights
- **Vendors/integrators:** hardware/software delivery, maintenance
- **Occupants:** feedback and behavior; demand flexibility

Collaboration Tools

RACI clarity | Shared glossary | Interface specs | Change control

Tool	Purpose
RACI clarity	Defines who's Responsible, Accountable, Consulted, and Informed.
Shared glossary	Ensures everyone uses common, consistent terminology.
Interface specs	Documents how systems and teams connect and communicate.
Change control	Manages updates and changes to avoid misalignment or conflict.

The T-Shaped Professional: Depth + Breadth



Balance of specialized expertise and cross-disciplinary skills

↓ Vertical (depth)

- Your primary specialty (e.g., HVAC controls, energy analytics)
- Evidence: portfolio piece, certification, project impact

→ Horizontal (breadth)

- Literacy in IT (networks, APIs), data (pandas/SQL), and ML basics
- Soft skills: problem framing, storytelling, collaboration

How to build your "T" in this subject

- Choose a project that leverages your depth and stretches your breadth
- Produce artifacts: clear diagrams, a runnable notebook, a concise briefing
- Reflect: what you learned, what you'd do differently next time

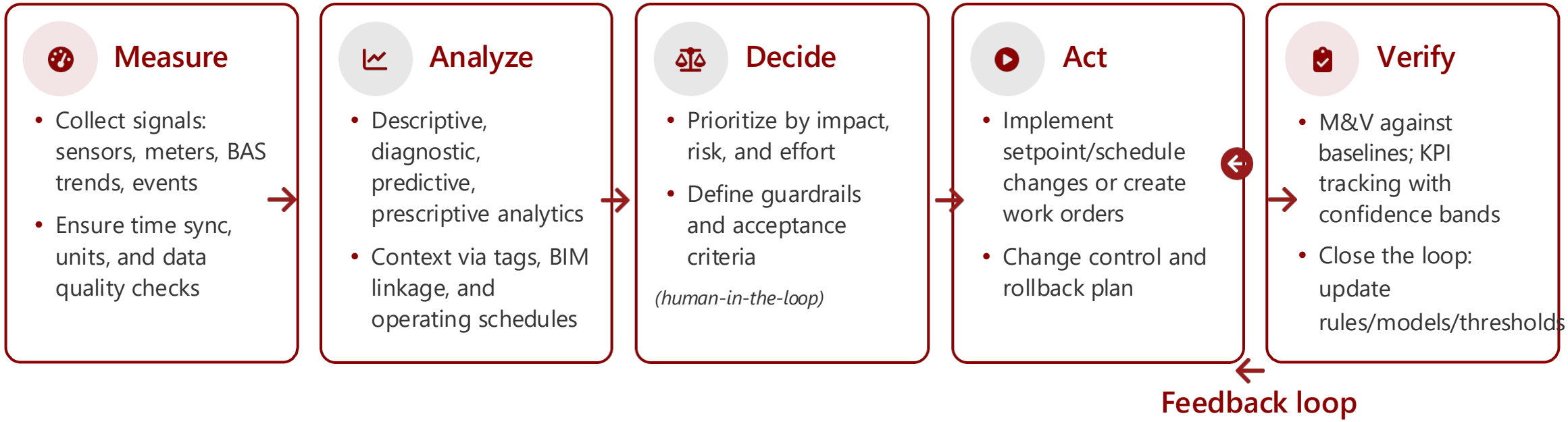
Career signal

Show outcomes tied to KPIs; demonstrate reproducibility and teamwork

Key Artifacts to Showcase Your T-Shape

System Diagram | Notebook | Briefing

Data-Driven Decision Loop: Measure → Analyze → Decide → Act → Verify



Use Case: Energy Optimization

Problem Focus

- Reduce energy and peak demand without violating comfort

Data Required

- Power/energy meters; HVAC points (SAT, valve/damper positions, flows)
- Weather (dry-bulb, humidity), occupancy schedules, tariffs

Methods

- Baseline modeling (regression or change-point)
- Predictive control or schedule optimization; rule tuning
- DR preparation: load shed strategies with pre-cooling

Actions

- Setpoint resets (SAT, chilled water), schedule adjustments, VFD optimization
- Demand response participation with playbooks

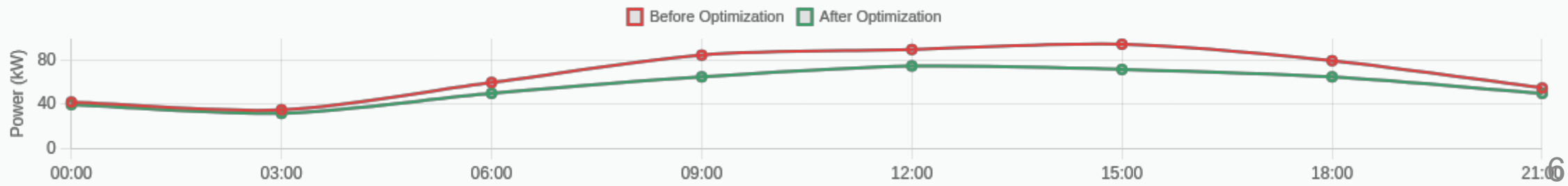
KPIs & Verification

- kWh, peak kW, comfort time-in-range; normalized vs. weather/occupancy
- M&V plan (pre/post or counterfactual)

Risks & Mitigations

- Comfort drift (set guardbands); equipment constraints (min/max limits)
- Rebound effects (staggered recovery)

Before/After Daily Energy Profile



Use Case: IEQ and Health

Problem Focus

Maintain comfort and air quality while managing energy

Data & Quality

- Sensor calibration and drift checks; placement strategy
- Occupancy context (schedules, counters, proxies)

Actions

- Demand-controlled ventilation; filter maintenance
- Humidity control; zonal balancing
- Ventilation scheduling aligned to usage

Risks & Ethics

- Privacy in occupancy sensing; transparency and data

Key Parameters

- Temperature bands, RH, CO_2, PM2.5
- VOCs, noise, light

Methods

- Threshold tracking; time-in-compliance metrics
- Correlate IEQ excursions with occupancy and HVAC modes
- Multi-objective optimization (energy vs. IEQ)

KPIs & Reporting

- % of occupied hours within IEQ targets
- Complaint rate vs. measured conditions

Use Case: Safety and Fault Detection & Diagnostics (FDD)

Safety Monitoring

- Thermal anomalies, leakage detection
- Smoke/CO alarms, access anomalies

Data Signals

- Temperatures, pressures, flows
- Currents, vibrations, door states
- Cameras (if authorized)

Performance

- Precision/recall trade-offs
- Alarm fatigue management
- Root cause trees and evidence snapshots

FDD Approaches

- Rule-based (thresholds, patterns)
- Model-based (physics)
- Data-driven (ML)

Workflow

Detect → Prioritize → Diagnose → Create work order → Verify

Integration

- Computerized Maintenance Management System (CMMS) ticketing
- SLA tracking
- Lessons learned to update rules

Ask: "Name one multi-sensor combination that increases confidence in a fault diagnosis."

Use Case: Space and Asset Management

Space Utilization

- Occupancy analytics (counters, BLE/Wi-Fi aggregates, booking data)
- Hot-desking policies; meeting room right-sizing

Asset Performance

- Runtime hours, start/stop counts, condition indicators (vibration, temp)
- Remaining useful life estimates; maintenance prioritization

Methods

- Utilization heatmaps; queuing or scheduling analysis
- Predictive maintenance models with simple features first

Actions

- Re-allocate space; adjust cleaning/maintenance schedules
- Proactive parts replacement; targeted inspections

KPIs

- Utilization % by zone/time; maintenance backlog
- Unplanned downtime; energy efficiency per use

Privacy and Ethics

- Aggregate/anonymous analytics; clear communication to occupants
- Data minimization; purpose limitation; consent practices

Prompt: "Which asset metric would you track first to predict maintenance needs, and why?"

Case Snapshot: Data-Driven Ventilation Win

Context

Mixed-use academic building with frequent CO_2 excursions and rising energy costs

Data Used

- CO_2, temperature, AHU supply airflow/temperature
- Occupancy schedules, weather forecasts

Method

- Baseline (pre/post with weather normalization)
- Demand-controlled ventilation (CO_2 targets)
- Supply temp reset with outdoor temp
- Occupancy-aligned schedules

Results

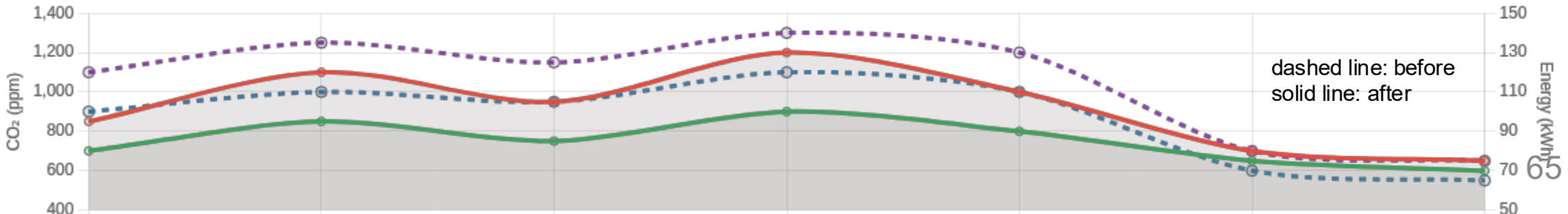
15%	60%	93%
HVAC Energy Reduction	CO_2 Excursion Reduction	Comfort Time-in-Range

Improved from 82% to 93% comfort time-in-range

Safeguards

- CO_2 cap at 900–1000 ppm
- Minimum outdoor air fraction
- Humidity guardbands
- Staged recovery after setbacks

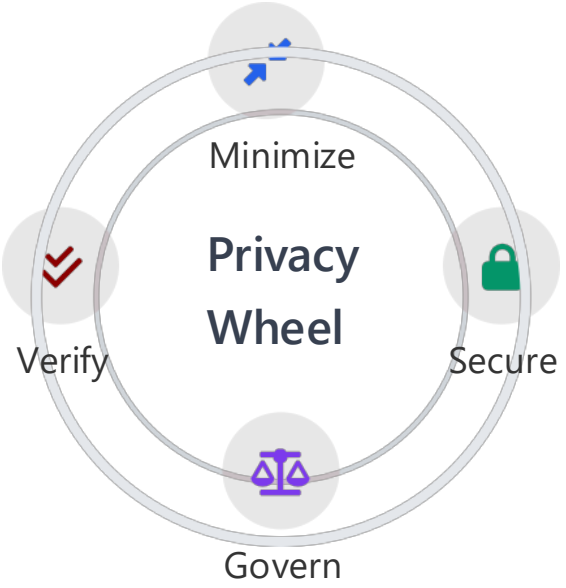
Weekly CO_2 & Energy Comparison



Ethics, Privacy, and Trust in Smart Buildings

Core Principles

- Data minimization and purpose limitation
- Transparency and informed notice; consent where applicable
- Anonymization/pseudonymization; avoid PII by default



High-Risk Categories

- Video/audio, biometrics, fine-grained location traces
- Use only with strong justification, safeguards, and approvals

Access & Security

- Role-/attribute-based access control; least privilege
- Secure transport (TLS), audit logs, key/cert management
- Retention and deletion policies aligned to purpose

Fairness & Bias

- Sensor placement bias; calibration disparities
- Inclusive design for occupants and staff

Governance

- Data Protection Impact Assessment (DPIA)
- Align with institutional/legal requirements
- Vendor due diligence and data safeguards

Ethics Scenarios: Quick Knowledge Check

Scenario A: Using WiFi MAC addresses to estimate occupancy

- Consider: hashing/salting to anonymize data
- Consider: opt-out mechanism for occupants
- Consider: retention limits (how long to store data)
- Consider: Data Protection Impact Assessment (DPIA)

Scenario B: Facial recognition for restricted lab access

- Consider: necessity vs. less intrusive badge alternatives
- Consider: bias and error rates across different demographics
- Consider: informed consent requirements
- Consider: secure storage and encryption of biometric data

Scenario C: Sharing building data with a third-party energy pilot

- Consider: data sharing agreement with clear scope/purpose
- Consider: re-use limitations and prohibitions

Transition: From Concepts to the Data Ecosystem

We have covered:

- Definitions and boundaries of Building Informatics
- Value chain from data to actionable insights
- Use cases across energy, IEQ, safety, and assets
- Ethics, privacy, and trust considerations

Next up:

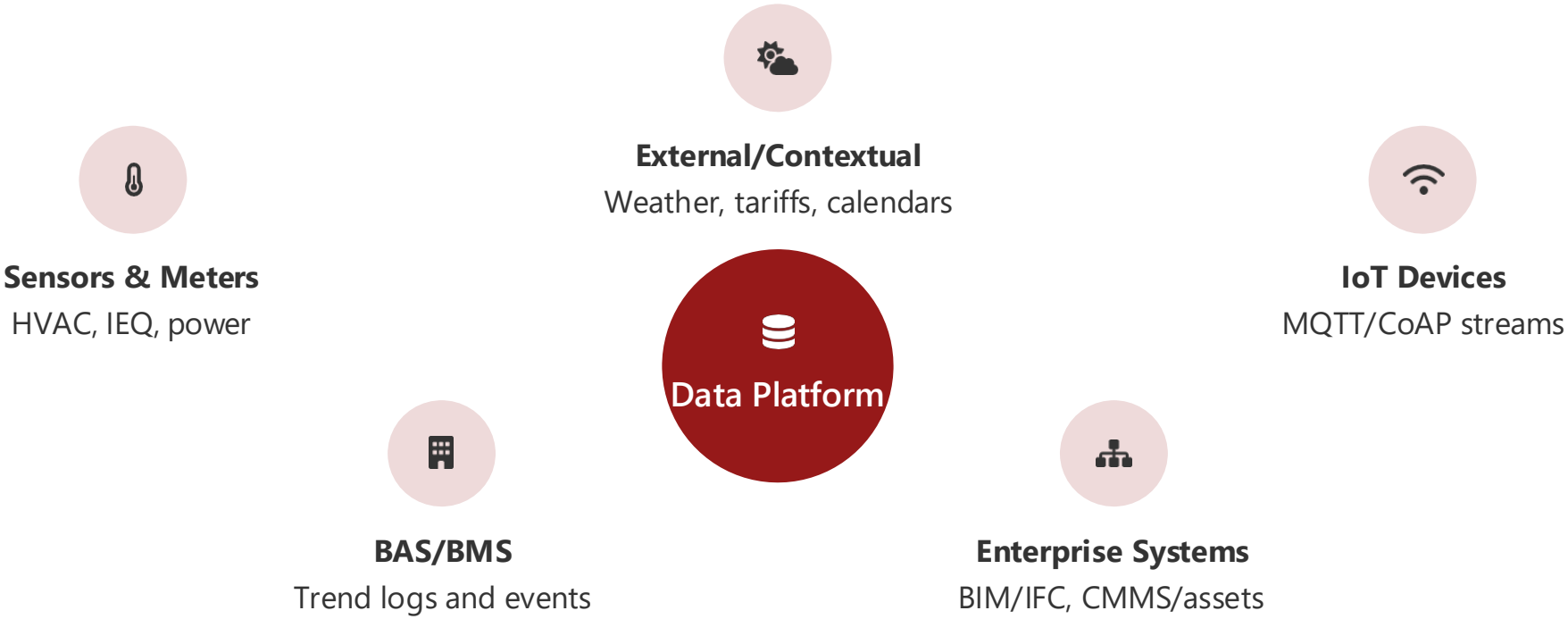
- Data sources in buildings (sensors, BAS, IoT, enterprise)
- Data types, quality, and characteristics
- Lifecycle: acquisition → processing → application



Learning objective:

Identify, characterize, and evaluate building data sources for analytics and control

Building Data Sources: The Landscape



Characteristics

- Cadence: sub-second to daily; latency tolerance
- Reliability and completeness; clock sync and time zones
- Ownership and access rights; API/export availability

Integration Considerations

- Naming/tagging consistency; ID mapping across systems
- Data volume/cost and storage tiering

Sensors in Buildings: Types, Accuracy, Placement

Types and Tech

- Thermal: temperature (thermistors, RTDs), RH
- Air quality: CO_2 (NDIR), PM2.5 (laser scattering), VOCs (MOX)
- Occupancy: PIR, ultrasonic, camera-based, BLE/Wi-Fi aggregates, CO_2 proxy
- Electrical/mechanical: CTs/power meters, pressure/flow, vibration

Data Quality

- Accuracy, precision, resolution, drift
- Calibration schedules and verification
- Sampling rates and filtering techniques
- Unit standards and consistency

Placement Guidelines

- Represent occupied zone; avoid drafts/sunlight/heat sources
- Height and setback standards (1.1–1.5 m for IEQ sensors)
- Consider spatial variations and representative coverage
- Avoid interference from nearby equipment or activities

Documentation

- Sensor ID, location, unit, calibration date
- Maintenance log and calibration history
- Manufacturer specs and expected accuracy
- Data verification procedures and reliability metrics

Quick design task: "For a 12-person meeting room, which occupancy sensor(s) would you choose and where would you place them? 30 seconds to think."

BAS/BMS as Data Sources: Points, Trends, and Events

Point Types

- Analog/Boolean inputs and outputs
- Key HVAC/lighting points (SAT, RAT, VFD speed)
- Valve/damper positions, zone temp
- Occupancy command status

Trend Logs

- Sampling intervals (1-15 min)
- Buffer depth and rollover behavior
- Alarms/events with priorities

Common Issues

- Inconsistent naming conventions
- Sparse or irregular logging periods
- Buffer overflows causing data loss
- Unit mismatches and conversions
- Clock drift between systems
- Point mapping errors across systems

External and Contextual Data: Weather, Tariffs, Schedules

Weather

- Dry-bulb, wet-bulb/dew point, RH, wind, solar irradiance
- Sources: on-site sensors vs. API; data cadence and gaps

Calendars & Schedules

- Academic/holiday calendars; occupancy schedules
- Room booking systems and utilization data

Integration Notes

- Time zones/DST alignment; units; data licensing
- Caching and retries for external APIs
- Use for forecasting, normalization, and rule conditions

Energy & Tariffs

- Time-of-Use rates; demand charges; real-time pricing
- DR event notices and participation windows

Space & Asset Context

- Room type, floor area, orientation
- Asset specifications and locations

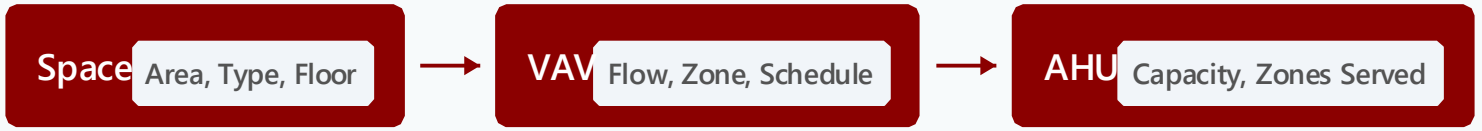
BIM as a Data Source: Objects, Properties, Relationships

What BIM provides

- Objects: spaces, zones, systems, equipment (e.g., chiller)
- Properties: room type, floor area, orientation, design airflows, setpoints
- Relationships: system-to-equipment-to-space connectivity

Useful exports for analytics

- Space list with area, use type, floor, zone
- Asset registry with IDs, specs, location, system membership
- Envelope data (U-values, SHGC), shading, orientation
- COBie/IFC tables for structured handover



Sample Table: Spaces

SpaceID	Area (m^2)	UseType	Floor
S-101	45.6	Classroom	1
S-102	18.2	Office	1

Time-Series Data: Granularity, Seasonality, and Alignment

Granularity and Cadence

- Regular vs. irregular sampling; 1s/1min/15min/hourly
- Anti-aliasing: match sampling to dynamics

Alignment

- Time zones and DST; store in UTC, display local
- Resampling (mean/sum/max), interpolation, window functions

Patterns

- Daily/weekly/seasonal cycles; holidays; change-points
- Lagged effects; cross-correlation with weather/occupancy

Quality

- Gaps, duplicates, out-of-order points; data latency/freshness
- Rolling stats; robust outlier detection; sanity bounds

Data Quality Dimensions and Metrics

Core dimensions

- Completeness, accuracy, precision, validity, consistency
- Timeliness/freshness, uniqueness, integrity, reliability
- Drift (sensor, process), coverage (spatial/temporal)

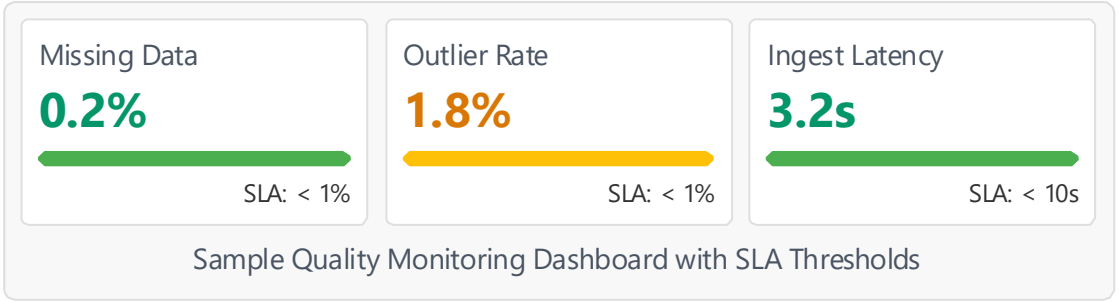
Governance

- Data quality SLAs; monitoring dashboards
- Automated tests at ingest (schema, ranges, units)
- Escalation playbooks for quality incidents

Quantifying quality

- % missing, duplicate rate, gap length distribution
- Outlier rate (z-score/IQR), spike/stuck metrics
- Latency (ingest delay), clock skew; unit conformity checks

Quality Dashboard Example



Sample Quality Monitoring Dashboard with SLA Thresholds

Common Data Issues and How to Handle Them

Typical Issues

- Missing data, gaps, duplicates, out-of-order timestamps
- Spikes/noise, stuck-at values, clipping/saturation
- Clock drift/DST errors; unit mismatches; sign inversions
- Mislabeling (RAT vs. SAT), sensor swaps, identifier churn
- Trend buffer rollover/compression artifacts

Quality Flag Example:

OK | MISSING | IMPUTED | SUSPECT

Flag data quality issues clearly rather than silently correcting

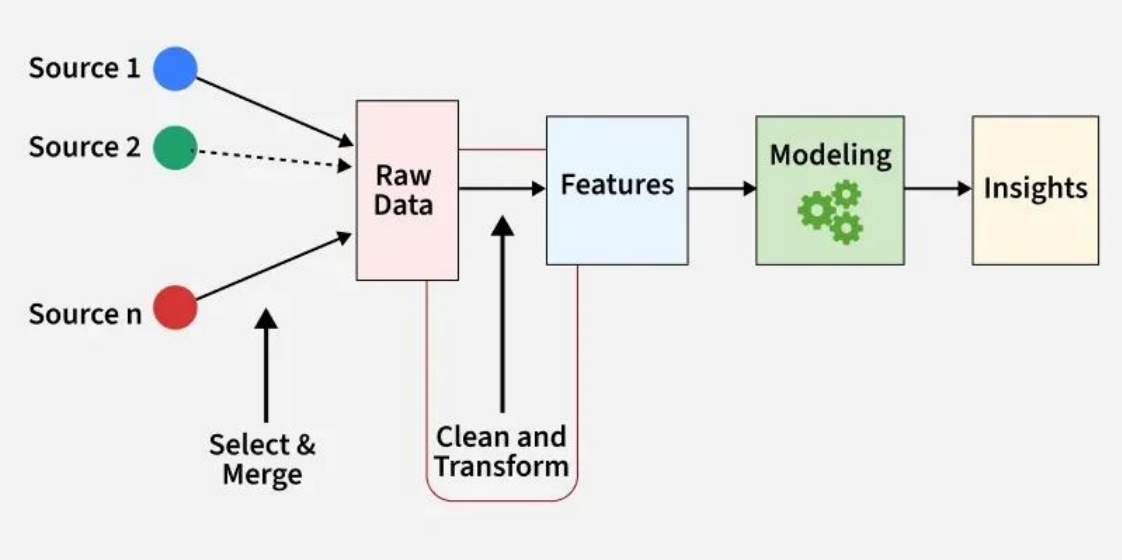
Mitigations

- Validation on ingest; de-duplication; sort by event-time
- Imputation with clear marking; avoid silent corrections
- Robust smoothing; spike filters with physical limits
- Enforce NTP; unit tests for units; reconciliation checks
- Change logs; mapping tables; controlled rename/migration

Issue → Mitigation Examples:

- **Time Drift** → NTP synchronization, timezone tests
- **Spikes/Noise** → Median filters, physical bounds
- **ID Changes** → Mapping tables, stable UUIDs
- **Missing Values** → Flag gaps, context-aware imputation

Feature Engineering for Building Data





Sorce: https://www.geeksforgeeks.org/machine-learning/what-is-feature-engineering/

Feature Dictionary Example

Feature Name	Definition	Unit	Window	Source
temp_delta_24h	Zone temperature change over 24h	°C	24h	BAS
rolling_kW_max	Maximum power consumption	kW	4h	Meter
valve_cmd_actual_diff	Difference between commanded and actual valve position	%	Real-time	BAS

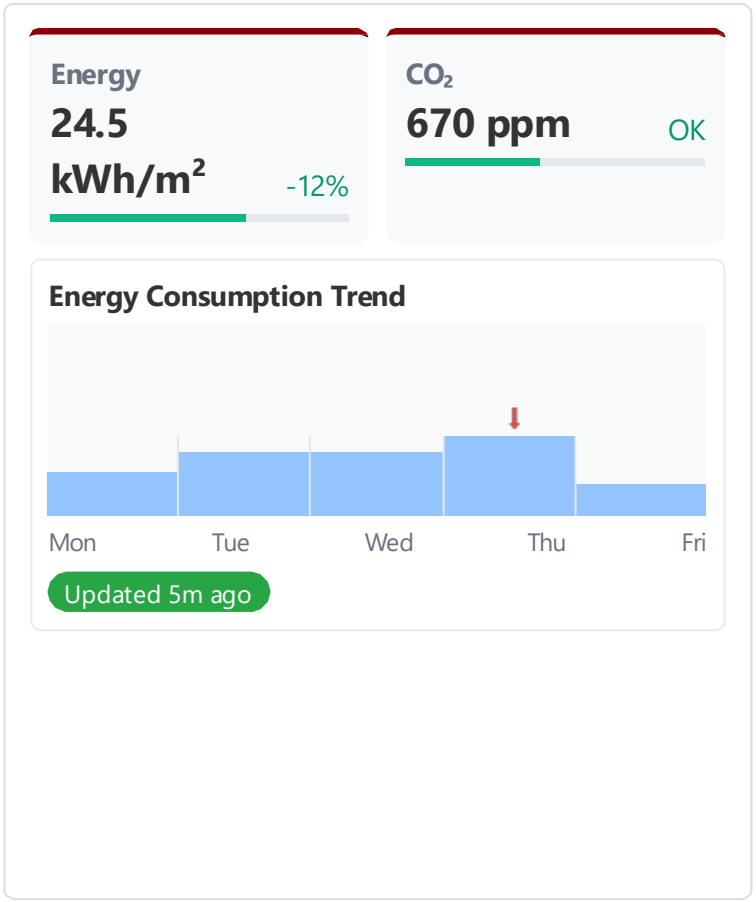
Dashboards and Monitoring: Designing for Decisions

Principles

- Audience-first (operator vs. manager)
- Less but better; consistent units
- Show status + trend + target
- Color with care; accessible design

Components

- KPI tiles with targets
- Time-series panels
- Drill-down filters
- Annotations (events/changes)



Freshness & Trust

- Data freshness indicator
- Quality flags
- Last change note

Anti-patterns

- 20+ KPIs on one page
- Mixed units/axes
- Unlabeled thresholds

Storytelling

- One-page narrative flow:
- Problem → Evidence → Action → Outcome

Digital Twins for Buildings

BIM can provide part of the static semantic/geometric foundation, while a **digital twin** links physical assets with synchronized operational data, models, and feedback. It goes beyond a simple 3D model, becoming a living, evolving replica of the building that integrates real-time data from IoT sensors, operational history, and predictive analytics.



https://www.autodesk.com/design-make/articles/digital-twin-examples

This dynamic, live model provides a comprehensive digital identity for the building and its systems, allowing for **real-time monitoring, in-depth analysis, simulation**, and prediction to improve efficiency, optimize operations, enhance sustainability, and support informed decision-making throughout the building's lifecycle.

Digital Twins in Practice: Operate, Simulate, Improve

Definition: A synchronized virtual representation of assets, systems, and spaces, linked to live data

Operational Twin

- Live KPIs by system/space
- Context-rich FDD with anomaly detection
- Asset histories and maintenance records
- Real-time alerts and operational dashboards

Simulation Twin

- Calibrated models for 'what-if' scenarios
- Schedule optimizations and retrofit evaluations
- Tariff and demand response simulations
- Training environment for operators

Integration Patterns

- BIM/IFC + tags + time-series
- ID governance across systems
- Change/version control for updates

Fidelity & Scope

- Model only where value is highest
- Keep rest descriptive/lightweight
- Balance complexity vs. usefulness

Performance & Sustainment

- Fast search; visual clarity; mobile access
- Data freshness SLAs; automated sync tests
- Clear content owner roles and responsibilities

Ask: "Name one scenario you would test in a simulation twin before implementing on a live building."

Digital Twin for The Hong Kong Polytechnic University



Developed by Guo Zhiling

Thanks
&
Any Question?